भारतीय प्रौद्योगिकी संस्थान मुंबई

INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

पवई / Powai, मुंबई / Mumbai 400 076

Official Transcript

Roll Number:	160010010	Academic Unit (B.Tech.):	Mechanical Engineering
Name of the Student:	ARUSH GUPTA	Academic Unit (M.Tech.):	Electrical Engineering
Programme:	Interdisciplinary Dual Degree (IDDD)	(Specialization)	(Microelectronics)
		Joining Month & Year:	July 2016

Academic Year: 2016 - 2017, Term: Semester Autumn

Code	Name	Credits	Tag	Grade/ Marks	Code	Name	Credits	Tag	Grade/ Marks
BB 101	Biology	6.0	MA	BB	MA 105	Calculus	8.0	MA	BB
CH 105	Organic & Inorganic Chemistry	4.0	MA	AB	ME 119	Engineering Graphics & Drawing	5.0	MA	AA
CH 107	Physical Chemistry	4.0	MA	BC	NOCS01	NCC/NSS/NSO	0.0	MA	PP
CH 117	Chemistry Lab	3.0	MA	AA	PH 107	Quantum Physics and application	6.0	MA	BB

SPI=8.44/10 CPI=8.44/10

Academic Year: 2016 - 2017, Term: Semester Spring

Code	Name	Credits	Tag	Grade/ Marks	Code	Name	Credits	Tag	Grade/ Marks
AE 102	Data Analysis and Interpretation	6.0	MA	AB	ME 113	Workshop Practice	4.0	MA	AA
AE 152	Introduction to Aerospace Engg.	6.0	MA	BB	NOCS02	NCC/NSS/NSO	0.0	MA	PP
CS 101	Computer Programming and Utilization	6.0	MA	AA	PH 108	Basics of Electricity & Magnetism	6.0	MA	AA
MA 106	Linear Algebra	4.0	MA	AA	PH 117	Physics Lab	3.0	MA	AB
MA 108	Differential Equations	4.0	MA	AA					

SPI=9.46/10 CPI=8.97/10

Academic Year: 2017 - 2018, Term: Semester Autumn

Code	Name	Credits	Tag	Grade/ Marks	Code	Name	Credits	Tag	Grade/ Marks
EE 101	Introduction to Electrical and Electronics Circuits	8.0	MA	AB	ME 209	Thermodynamics	6.0	MA	AB
					ME 219	Fluid Mechanics	8.0	MA	BB
HS 101	Economics	6.0	MA	AA	MM 207	Engineering Metallurgy	6.0	MA	AB
ME 201	Solid Mechanics	6.0	MA	AB					

SPI=8.95/10 CPI=8.97/10

Academic Year: 2017 - 2018, Term: Semester Spring

Code	Name	Credits	Tag	Grade/ Marks	Code	Name	Credits	Tag	Grade/ Marks
CE 102	Engineering Mechanics	6.0	MA	AA	ME 213	Manufacturing Practice Lab	5.0	MA	AA
MA 214	Introduction to Numerical Analysis	8.0	MA	BC	ME 218	Solid Mechanics Lab	3.0	MA	AB
ME 202	Strength of Materials	6.0	MA	BC	ME 224	Fluid Mechanics Lab	3.0	MA	AA
ME 206	Manufacturing Processes I	6.0	MA	BC	ME 226	Mechanical Measurements	6.0	MA	AB

SPI=8.40/10 CPI=8.81/10

Academic Year: 2018 - 2019, Term: Semester Autumn

Code	Name	Credits	Tag	Grade/ Marks	Code	Name	Credits	Tag	Grade/ Marks
EP 332	Thermal and Statistical Physics	6.0	MA	AA	ME 346	Heat Transfer II	6.0	MA	AB
HS 307	Sociology	6.0	MA	AB	ME 374	Manufacturing Processes Lab	3.0	MA	BB
ME 307	Mechanical Measurements Lab	3.0	MA	AA	ME 661	Advanced Thermodynamics & Combustion	6.0	MA	AA
ME 311	Microprocessors and Automatic Control	6.0	MA	AA					
ME 338	Manufacturing Processes II	6.0	MA	BC					

SPI=9.14/10 CPI=8.88/10

Academic Year: 2018 - 2019, Term: Semester Spring

Code	Name	Credits	Tag	Grade/ Marks	Code	Name	Credits	Tag	Grade/ Marks
EE 755	Quantum Transport in Nanoscale Devices	6.0	AL	W	ME 308	Industrial Engineering and Operations Research I	6.0	MA	BB
EP 502	Superconductivity and Low Temperature Physics	6.0	AL	W	ME 310	Microprocessors and Automatic Control Lab	3.0	MA	BB
ES 200	Environmental Studies: Science and Engineering	3.0	MA	BB	ME 316	Kinematics and Dynamics of Machines	6.0	MA	AB
					ME 370	Kinematics and Dynamics of Machines Lab	3.0	MA	AB
HS 200	Environment	3.0	MA	AA	ME 372	Heat Transfer and Metrology Lab	3.0	MA	AB
ME 306	Applied Thermodynamics	6.0	MA	AB					

SPI=8.73/10 CPI=8.86/10

Signature Not Verified
Digitally signed by PREMKUMAR RATNAM
Date: 2021.10.18 05:57:32 +05:30
Reason: IIT BOMBAY NAD
Location: IIT BOMBAY

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INDIAN INSTITUTE OF TECHNOLOGY BOMBAY

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Name of the Student: ARUSH GUPTA | Roll Number : 160010010

Academic Year: 2019 - 2020, Term: Semester Autumn

Code	Name	Credits	Tag	Grade/ Marks
ME 423	Machine Design	8.0	MA	BB
ME 441	Applied Thermodynamics Lab	3.0	MA	BB
ME6101	Lattice Dynamics and Thermal Energy Transport	6.0	MA	AA
ME 651	Fluid Dynamics	6.0	MA	BC
ME 704	Computational Methods in Thermal & Fluid Engg	6.0	AL	BC
ME 743	Optical Methods in Mechanical Engineering	6.0	MA	AB

SPI=8.41/10 CPI=8.81/10

Academic Year: 2019 - 2020, Term: Semester Spring

Code	Name	Credits	Tag	Grade/ Marks
EE 451	Supervised Research Exposition	6.0	AL	AA
EE 727	Physics of Nanoscale Devices - II	8.0	MA	AB
ME 790	Materials modelling using atomistic first-principles calculations	6.0	MA	AB
PH 202	Waves & Oscillations & Optics	6.0	AL	W
PH 204	Quantum Mechanics I	8.0	AL	BB
PH 352	Introduction to Condensed Matter Physics	6.0	AL	AA
PH 438	Statistical Physics	6.0	MA	PP
PH 534	Quantum Information and Computing	6.0	AL	PP

SPI=9.00/10 CPI=8.82/10

Academic Year: 2020 - 2021, Term: Semester Project*

Code	Name	Credits	Tag	Grade/ Marks
EE 593	Dual Degree Project I	36.0	PR	AB

Academic Year: 2020 - 2021, Term: Semester Autumn

Code	Name	Credits	Tag	Grade/ Marks
EE 669	VLSI Technology	6.0	MA	CD
EE 733	Solid State Devices	6.0	MA	BC
EE 787	Topological Electronics	6.0	MA	CC

SPI=6.00/10 CPI=8.68/10

Academic Year: 2020 - 2021, Term: Semester Summer

Code	Name	Credits	Tag	Grade/ Marks
EE 594	Dual Degree Project II	36.0	PR	AA

- CPI (Project) = 9.50/10

Mandatory Course Credits (MA)	**= 294.0**	**CPI (Courses)**	**= 8.65/10**
Project Credits (PR)	**= 72.0**		
Net Mandatory Credits (MA+PR)	**= 366.0**	**CPI (Overall)**	**= 8.81/10**
Overall Completed Credits	**= 366.0**		
Overall Grade Points	**= 3226.0**		

Final Result

The student has completed the academic requirements of the programme in the month of July 2021 for the award of Bachelor of Technology in Mechanical Engineering and Master of Technology in Electrical Engineering with Specialization in Microelectronics(Without Honors)

Signature & Seal of Transcript Issuing Authority:

Joint/Assistant Registrar (Academic), IIT Bombay

Date: 07-October-2021

Place: Mumbai

CONTINUED

Official Transcript

Name of the Student: ARUSH GUPTA | Roll Number: 160010010

General Information

The medium of instruction at the Institute is English.
Course credits and grade: Each academic course is associated with a credit which is an indicator of its relative academic weight in calculating the academic performance. A two-letter grade is awarded to students on the basis of their performance in examinations and assignments of a specific course. The letter grades have numerical equivalents on a 0-10 scale as given below.

Letter Grade	AP	AA	AB	BB	BC	CC	CD	DD	FF	FR	W	DX	PP	NP	AU
Numerical Equivalent	10	10	9	8	7	6	5	4	0	0	–	–	–	–	–

FF: Fail, FR: Fail and repeat, W: Withdrawn, DX: Insufficient attendance, AU: Satisfactory performance in an audit course, PP: Pass, NP: Not Pass. The minimum passing grade in a course is DD. The grade AP is awarded to students with exceptional performance in core courses of a programme. Numerical equivalents of letter grades are referred to as grade points.
The numerical grade points are not convertible into marks or percentages.
Performance Indicators: The performance of a student in a semester is given by a number called the Semester Performance Index (SPI),which is the weighted average of the earned grade points in the courses during the semester.
If a student has courses with credits $C_1, C_2, \ldots, C_n$, with grade points of $G_1, G_2, \ldots, G_n$ respectively, then

Semester Credits = $C_1 + C_2 + \ldots + C_n$.	Semester Grade Points = $C_1G_1 + C_2G_2 + \ldots + C_nG_n$.	SPI = Semester Grade Points ÷ Semester Credits.

Cumulative Performance Index (CPI) is the weighted average of the grade points in the courses in all semesters. The indices SPI and CPI are calculated upto two decimal places.
*The project grades are included for calculation of overall CPI which is shown in the CPI (Overall).
Courses are tagged as MA: Mandatory (Core/Elective), MI: Minor, HO: Honours, AL: Additional Learning, AU: Audit

- Each degree programme has mandatory credits consisting of core courses, elective courses, and non credit courses. These courses are tagged as MA.
- For calculation of SPI and CPI, grades obtained only in mandatory courses (MA) are considered.
- Students can supplement the learning experience by crediting additional courses. Credits earned in these courses, when appropriate, can earn additional credentials either in the form of "Honours" (HO) in the chosen discipline or "Minor" (MI) in another discipline or both.
- "Honours" is not indicative of proficiency, and can be earned by completing the additional prescribed set of advanced core and elective courses in the chosen discipline. "Minor" can be earned by completing the prescribed set of courses in a discipline other than the chosen discipline. Additional courses that are not used for earning "Honours" or "Minor" are tagged as "Additional Learning" (AL).
- The AU is awarded based on satisfactory attendance and fulfilling the minimum requirements as set by the course instructor.It carries no grade points and does not figure in SPI or CPI calculations.
- PP or NP is awarded in some credit courses that are not earmarked with a letter grade. Correspondingly, PP/NP does not carry a grade point.
- O-IITB is/are the Course(s) completed by a student outside IIT Bombay (NPTEL/ Swayam). These course(s) contribute towards the completion of credits for a degree requirement. However, grades/marks earned for such course(s) is/are not considered for SPI / CPI calculation.

The Institute does not award any class or division. Notionally, the CPI may be multiplied by a factor of 10 to obtain a numerical percentage.

The veracity of this document can be ascertained by using the verification ticket number in the URL given at the bottom of this page.

END OF TRANSCRIPT | Roll Number: 160010010

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